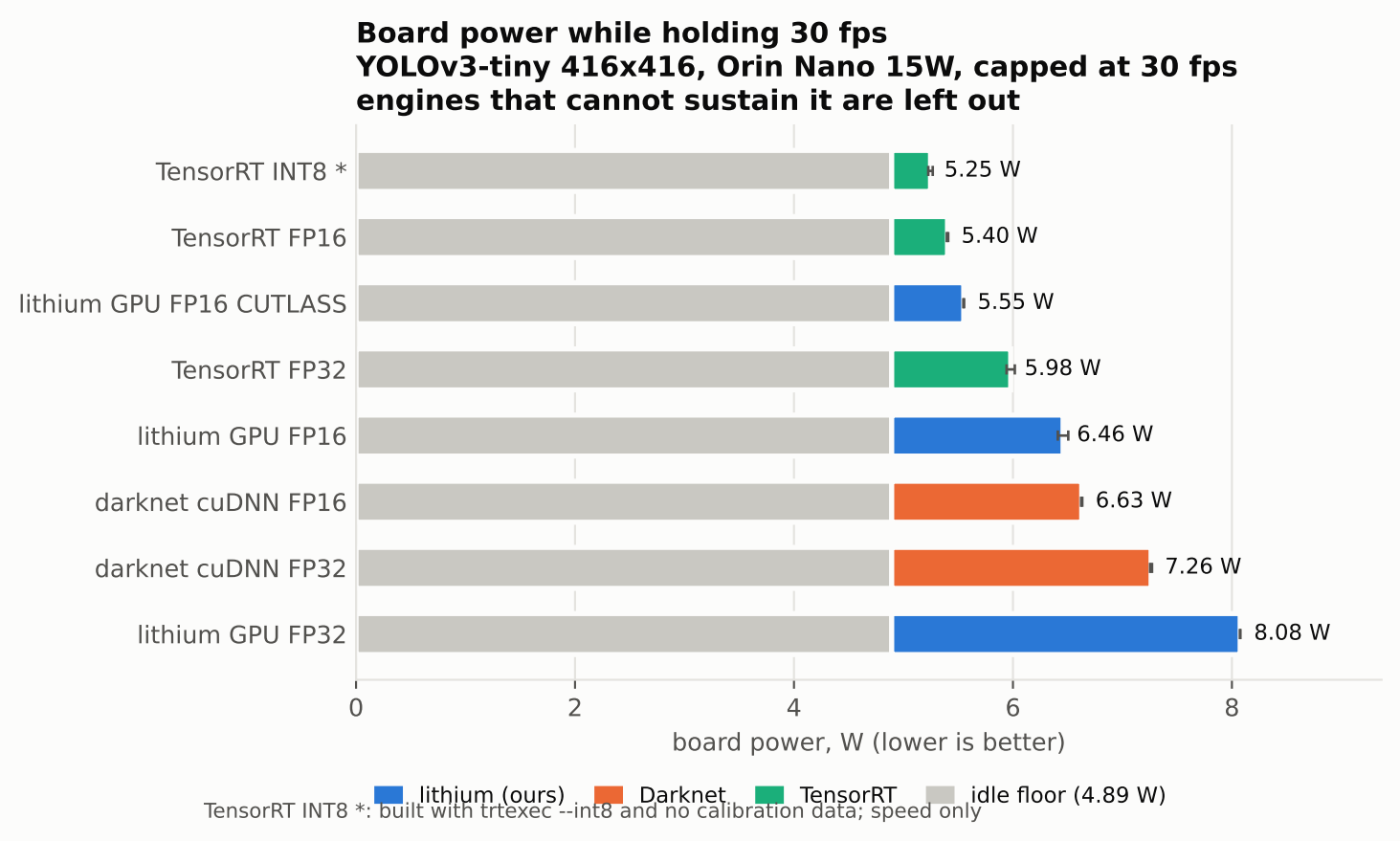
Board power while holding 30 fps
YOLOv3-tiny 416x416, Orin Nano 15W, capped at 30 fps
engines that cannot sustain it are left out
TensorRT INT8 *
TensorRT FP16
lithium GPU FP16 CUTLASS
TensorRT FP32
lithium GPU FP16
darknet cuDNN FP16
darknet cuDNN FP32
lithium GPU FP32
5.25 W
5.40 W
5.55 W
5.98 W
6.46 W
6.63 W
7.26 W
8.08 W
0
2
4
6
8
board power, W (lower is better)
lithium (ours)
Darknet
TensorRT
idle floor (4.89 W)
TensorRT INT8 *: built with trtexec --int8 and no calibration data; speed only